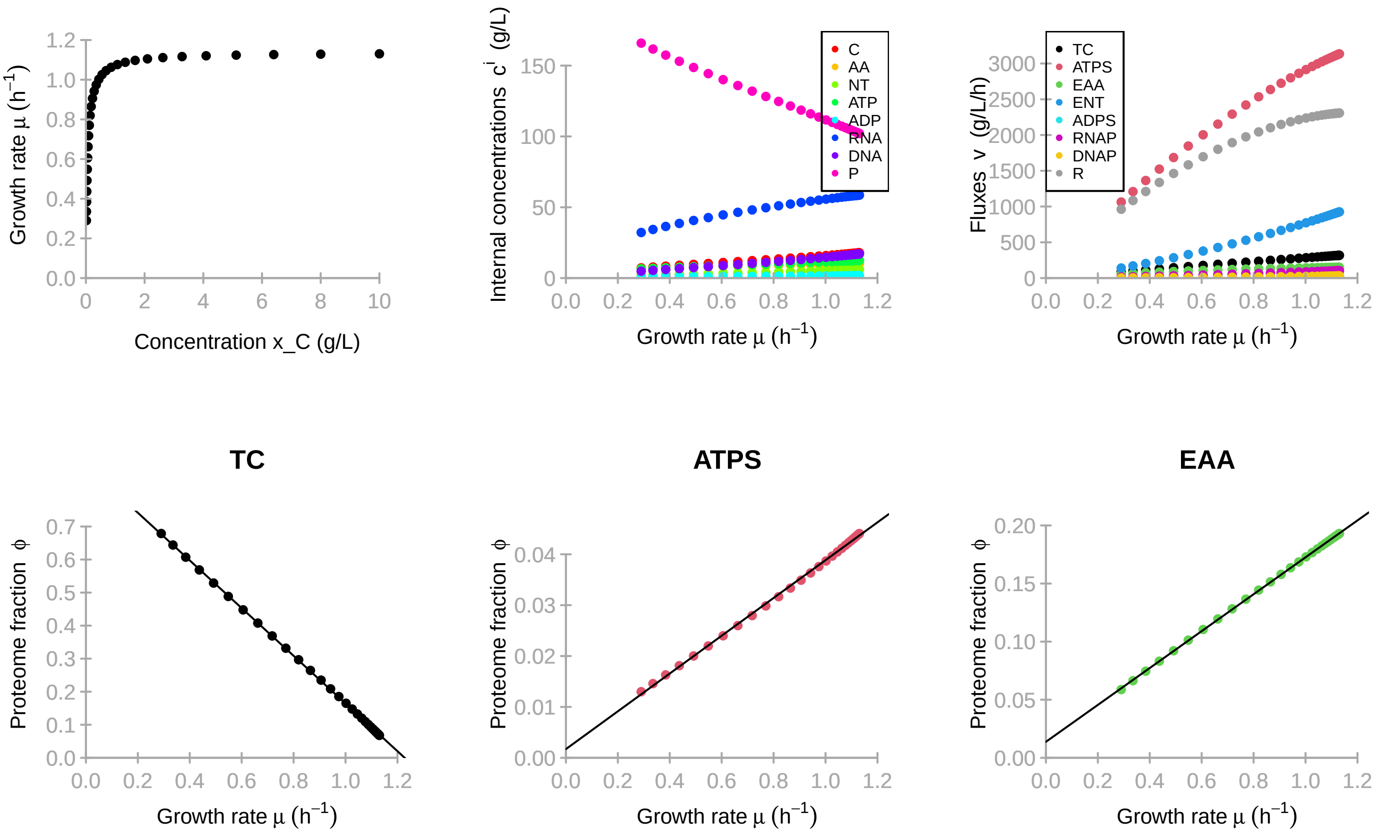
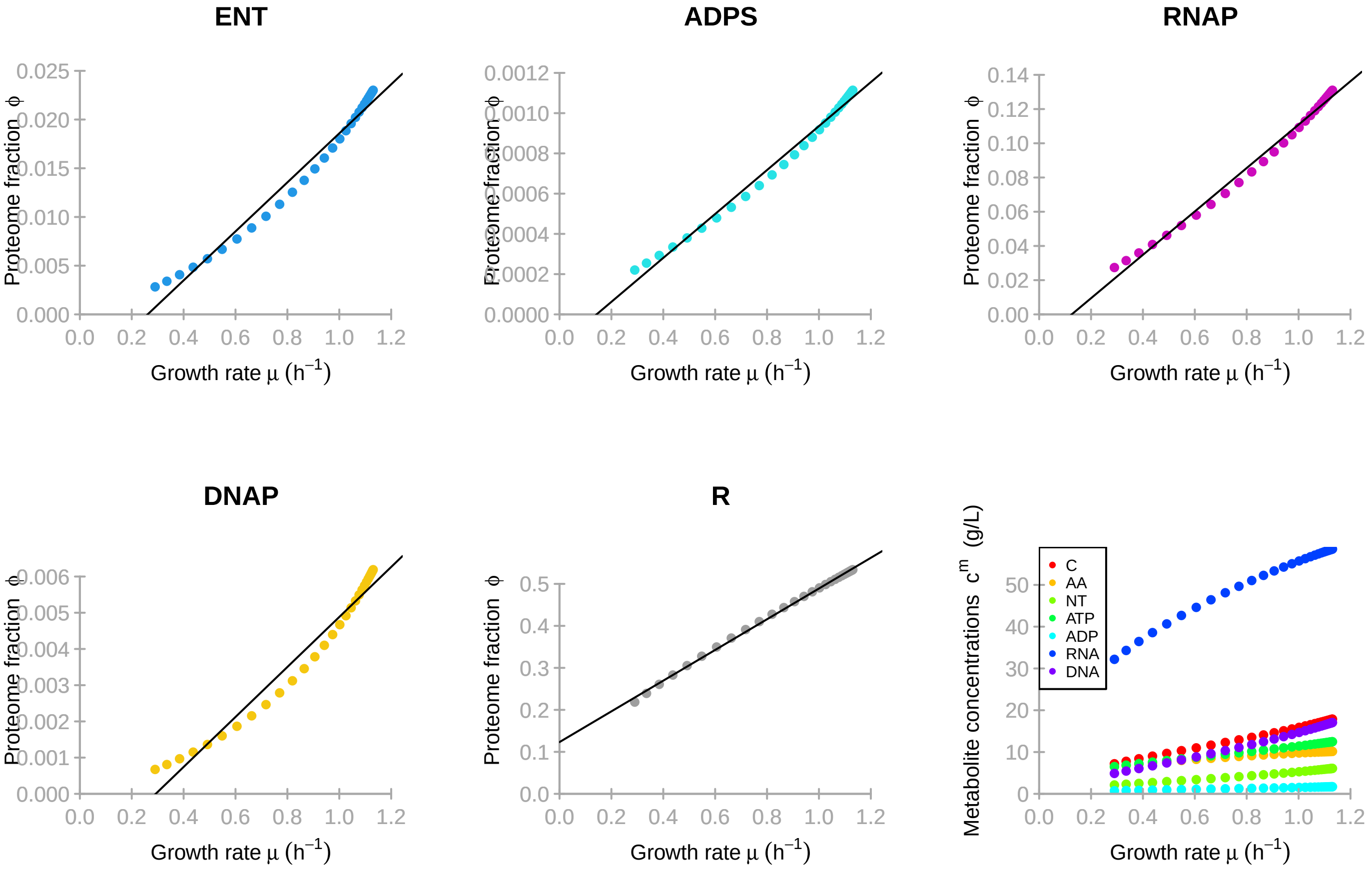
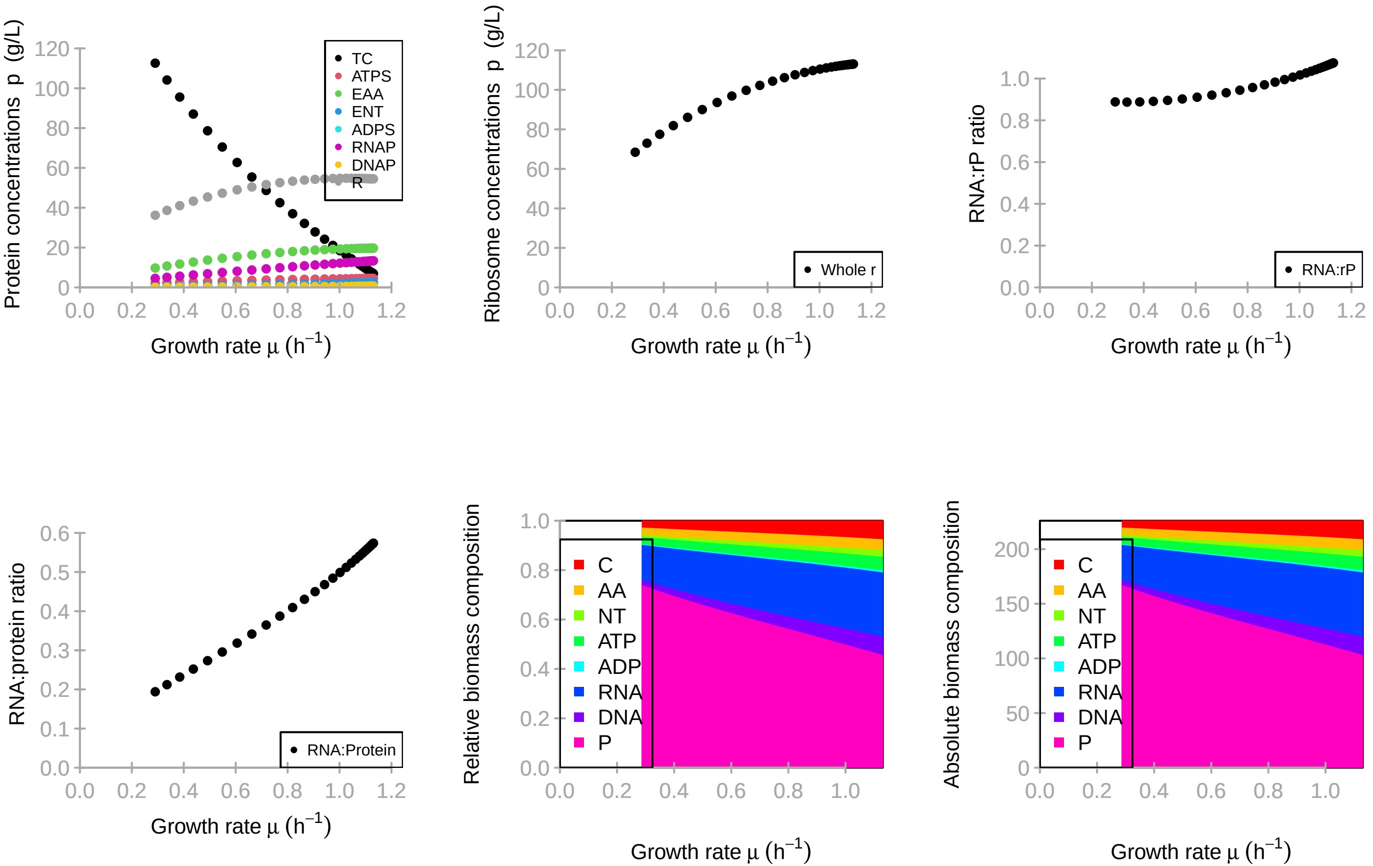
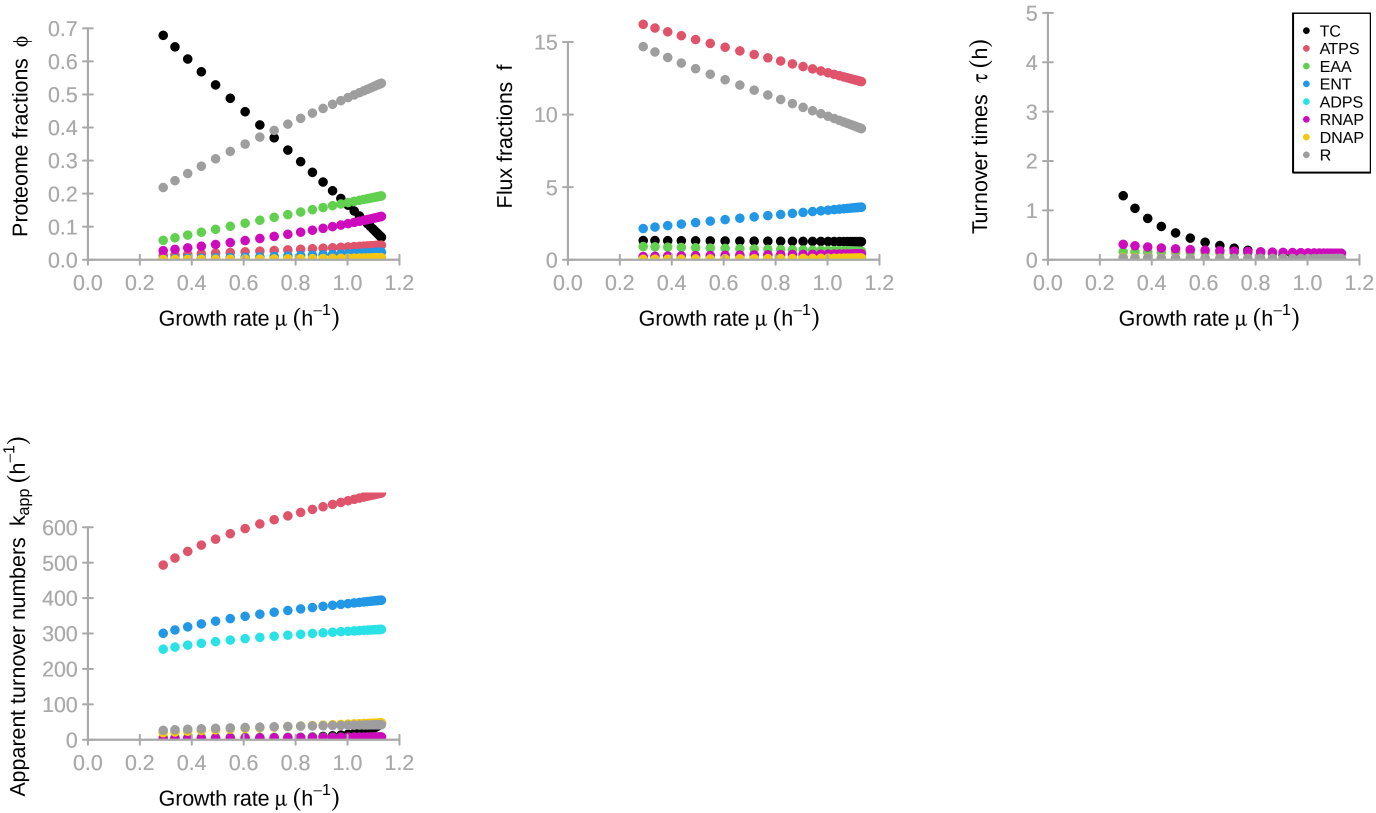


Growth rate μ (h^−1)
Concentration x_C (g/L)
Internal concentrations c^i (g/L)
Growth rate μ (h^−1)
C
AA
NT
ATP
ADP
RNA
DNA
P
Fluxes v (g/L/h)
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
TC
ATPS
EAA
Proteome fraction ϕ


ENT
ADPS
RNAP
DNAP
R
Proteome fraction φ
Growth rate μ (h⁻¹)
Metabolite concentrations c^m (g/L)
C
AA
NT
ATP
ADP
RNA
DNA
0.025
0.020
0.015
0.010
0.005
0.000
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.006
0.005
0.004
0.003
0.002
0.001
0.000
0.5
0.4
0.3
0.2
0.1
0.0
50
40
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
1.2


Protein concentrations p (g/L)
Growth rate μ (h−1)
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
Ribosome concentrations p (g/L)
Whole r
RNA:rP ratio
RNA:rP
RNA:protein ratio
RNA:Protein
Relative biomass composition
Absolute biomass composition
C
AA
NT
ATP
ADP
RNA
DNA
P


Proteome fractions ϕ
Growth rate μ (h⁻¹)
Flux fractions f
Turnover times τ (h)
TC
ATPS
EAA
ENT
ADPS
RNAP
DNAP
R
Apparent turnover numbers k_app (h⁻¹)
0.0 0.2 0.4 0.6 0.8 1.0 1.2
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
0 5 10 15
0 1 2 3 4 5
0 100 200 300 400 500 600

M

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
C	1	−0.17	−1	−0.04	0.13	0.36	0.36	0.16
AA	0	0	1	0	0	0	0	−0.06
NT	0	0	0	0.19	−1	−1	−1	0
ATP	0	0.98	0	−0.96	0	0	0	−0.94
ADP	0	−0.83	0	0.81	0.87	0	0	0.79
RNA	0	0	0	0	0	0.64	0	0
DNA	0	0	0	0	0	0	0.64	0
P	0	0	0	0	0	0	0	0.05

K

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
x_C	1	0	0	0	0	0	0	0
x_W	0	1	0	0	0	0	0	0
C	10.6	3.5	3.5	3.5	10.6	0	0	0
AA	0	0	2.3	0	0	0	0	0.8
NT	0	0	0	2.3	0.8	0.8	0.8	0
ATP	0	2.3	0	0.8	0	0	0	0.8
ADP	0	0.2	0	0.7	0.7	0	0	0
RNA	0	0	0	0	0	0	0	0
DNA	0	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0

KA

	TC	ATPS	EAA	ENT	ADPS	RNAP	DNAP	R
x_C	0	0	0	0	0	0	0	0
x_W	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0	0
AA	0	0	0	0	0	0	0	0
NT	0	0	0	0	0	0	0	0
ATP	0	0	0	0	0	0	0	0
ADP	0	0	0	0	0	0	0	0
RNA	0	0	0	0	0	0	0	72
DNA	0	0	0	0	0	11	11	0
P	0	0	0	0	0	0	0	0

kcat

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]
kcatf	50.6	931.1	9.1	501.7	352.7	14.4	88.8	108.4
kcatb	0	0	0	0	0	0	0	0

Keq

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]	[,8]
[1,]	Inf	Inf	Inf	Inf	Inf	Inf	Inf	Inf